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In AS Mathematics (8MA0)

Paper 01 Pure Mathematics

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General Comments

This paper provided opportunities for learners across the ability range to demonstrate their understanding of the AS Pure Mathematics content, with many learners making a confident start on the earlier questions and showing secure knowledge of core techniques such as algebraic manipulation, binomial expansion, differentiation, integration and the application of standard mathematical models. More demanding questions towards the end of the paper successfully differentiated between learners, particularly where problem solving, interpretation and proof were required.

Across the paper, learners were generally successful when carrying out routine processes but were less secure when required to combine different areas of mathematics, interpret the demands of a problem or justify their reasoning. Centres are encouraged to continue emphasising the importance of reading questions carefully, particularly where exact values, specific forms of answer or calculator restrictions are stated. Working should be clearly set out, with sufficient intermediate steps shown, especially in questions involving algebraic manipulation, logarithms, exponentials and proof.

A recurring theme was that some learners unnecessarily complicated otherwise accessible questions by using methods that increased the likelihood of error. Learners would benefit from greater confidence in selecting efficient approaches, checking the reasonableness of answers and drawing supporting diagrams or sketches where appropriate. Stronger responses were characterised by clear mathematical communication, accurate notation and careful use of given information throughout the question.

Comments on Individual Questions

Question 1

This question assessed learners' ability to solve inequalities and combinations of inequalities.

Part (a)

A large majority of learners clearly demonstrated their understanding in part (a). Where learners were not successful, the most common error was on either the expansion of the $4(x-3)$ (usually to $4x-3$) or the rearrangement. A small number of learners reversed the inequality to achieve $x <$ ... rather than $x >$ ... (while some instead gave the answer as $x =$...) Some learners also rounded the answer to 3.3 or 3.33, thus not gaining the accuracy mark. It was quite common for learners to state $x > \frac{20}{6}$ which also gained the full marks despite being unsimplified.

Part (b)

Part (b) was usually attempted but was less well answered. Where learners recognised the solutions directly from the given question, they gained the first M1 mark and often followed on with the correct inequality. A significant proportion of learners unnecessarily expanded the brackets and rearranged to try to solve again. This was often unsuccessful due to errors when rearranging and usually scored no marks. Where the correct critical values were achieved, a significant number of learners either did not attempt the inequality or stated the wrong inequality. The most common was $x < -\frac{5}{2}, x < 8$, which did not gain the accuracy mark. Many learners used the correct inequalities, however left the inequalities separate and did not combine them correctly, without including the word "and" or the intersection symbol. Where the solution was correctly achieved, a significant proportion of learners had drawn a quadratic graph which helped to identify the correct region.

Part (c)

Part (c) proved more demanding. However, many learners who had scored the previous four marks were able to correctly identify the combined solution. Some learners who had written down the wrong inequality in (b) were able to score this mark (due to their answers to part (a) and (b) being in the required format). Some learners simply listed the two answers to (a) and (b) without attempting to combine them, while others gave the answer to (b) as the answer to this question.

Question 2

This question was a standard question assessing binomial expansion and helped to give learners confidence in the early stages of the paper.

Part (a)

Part (a) was generally very well answered, and learners were able to demonstrate their understanding of binomial expansion. The most common errors were:

- misaligning the binomial coefficients with the correct powers of 2 and x
- not applying the power to the coefficient of x in the third and fourth terms, most often due to missing brackets in these terms

The majority of learners attempted the first four terms in the binomial expansion as required by the demand of the question, but some only provided three terms which meant they could not access the final mark in this part. Occasionally learners further manipulated the expression by dividing terms by a common factor, and this was penalised in the final mark.

Part (b)

For part (b), -0.01 was often stated with no working out, with learners deducing this directly from the given information. For those setting up and solving an equation, it was not uncommon to see sign slips and/or errors in place value in their working. A minority of learners left part (b) blank, however non-attempts were very rare with this question.

Question 3

This question on factorisation and solution of exponential equations was attempted by the vast majority of learners. It provided a reasonably good opportunity to differentiate between learners, with part (b) proving more challenging for less confident learners.

Part (a)

Part (a) was attempted well by most learners who took out a factor of x to obtain a quadratic which was then usually factorised correctly. The coefficient of 8 on the x^2 term did cause an issue for some, and it was clear that a number of learners solved the quadratic using either the quadratic formula, or their calculator solver, to find the roots before reverting to a factorised form. However, some learners struggled to deal with the root of $\frac{1}{8}$ and incorrect factorisations such as $\left(x-\frac{1}{8}\right)(x+4)$ were sometimes seen. Other learners attempted to take out a factor of $8x$ but then struggled to factorise the resulting quadratic which had fractional coefficients. Sign errors in the factorisation were sometimes seen, resulting in quadratic factorisation such as $(8x+1)(x-4)$. There were some learners who were able to carry out the factorisation correctly but did not gain the final mark as they did not bring all three factors together on a single line.

Part (b)

Part (b) was attempted by many but only about half were able to do this correctly. Many learners either did not recognise the connection with part (a) or did not know how to use what they had done in (a) to solve (b) and did not recognise that y was related to x from earlier in the question, even if they were able to manipulate the given equation to write $64^y = 4^{3y}$ and/or $16^y = 4^{2y}$. Those learners who spotted that $4^y = x$ were usually able to set up at least one equation and attempt to solve it. Although some learners were unable to make any further progress beyond the establishment of the equation(s), most learners who recognised the relationship between x and y wrote down all three equations and then, usually, rejected the two equations which had no real solutions.

In solving the equation $4^y = \frac{1}{8}$, a few different approaches were seen. Some used logs to solve the equation, others rewrote both sides of the equation as powers of 2 explicitly and others were able to obtain the solution by inspection. All three methods were acceptable. It was not acceptable, however, for the answer of $y = -\frac{3}{2}$ to appear without the intermediate step of $4^y = 8$ as this was an indication of reliance on using an equation solver on a calculator. Sometimes learners wrote down additional incorrect solutions such as $y = 0$ arising as an invalid solution to $4^y = 0$ and this did not gain the final accuracy mark unless rejected.

Question 4

This question, assessing the formulation of a model was very accessible and almost all learners made an attempt.

Part (a)

Part (a) was generally well answered by learners who recognised that the given model should be used directly. Most successful responses involved substituting the two pairs of values into the equation to form a pair of simultaneous equations and then solving for the unknown constants. However, a significant minority of learners attempted to construct a linear model from the original data, often calculating a gradient using the values of t rather than $\sqrt{t}$. Some learners didn't write the full model, just giving the values of a and b. Where simultaneous equations were formed correctly, some learners struggled to solve them accurately, while a small number also complicated their work unnecessarily by treating $\sqrt{4}$ and $\sqrt{9}$ as surds rather than simplifying them to numerical values.

Part (b)

Part (b) was answered correctly by many learners. The most common errors were not doubling the constant term and omitting the required units from the final answer.

Part (c)

Part (c) proved to be the most challenging part of the question. Many learners attempted to describe limitations of the model in general terms, often referring to volume being negative or simply stating that time cannot be negative, rather than using the mathematics of the model to determine a limiting value. Some learners correctly solved the relevant equation but rounded their answer prematurely, while others gave nearby integer values such as 13 or 14 rather than the exact value. Trial and improvement approaches were also common and were generally unsuccessful and did not gain credit.

Question 5

This question assessed learners' understanding of non-right-angled trigonometry and almost all learners made an attempt at both parts. A significant proportion of learners did not fully meet the requirement to give the area as a simplified surd in part (b) and adhere to the calculator warning, and as such were limited to 1 mark in part (b).

Part (a)

In part (a), the majority of learners used the cosine rule correctly and arrived at the given answer with no errors scoring both marks. Use of $\cos A$ was condoned for both marks in this question, but centres should advise learners to be mindful of using the correct notation used in the given answer. A few learners used the correct formula but made an error when simplifying such as $41 - 40\cos\theta = \cos\theta$ or reached the answer of $\frac{1}{5}$ not $-\frac{1}{5}$. It was not uncommon to see an incorrect cosine rule formula quoted.

Part (b)

In part (b), the majority of learners used their calculator to find the angle θ to be 101.5 and then used $\frac{1}{2}(4)(5)\sin 101.5°$ to find the correct but inexact area of the triangle. This use of calculator technology limited such responses to score only the second method mark. A much smaller proportion realised they needed to find an exact value for $\sin\theta$ first. Those that did usually used $\sin^2\theta + \cos^2\theta = 1$ and proceeded to find the correct exact answer for the area. Drawing a right-angled triangle and using Pythagoras to deduce that $\sin\theta = \sqrt{\frac{24}{25}}$ was seen less often but was usually correct if attempted. A very small number of learners made no attempt at this part.

Question 6

This question assessed learners' understanding of vectors and their ability to problem solve in order to solve an equation involving the magnitude of two vectors in part (b).

Part (a)

Part (a) was a very accessible question part with most learners successfully obtaining $p = -15$ In most cases, a scale factor between the two vectors was found, with some learners using the ratios of components instead. Some learners used a more complex approach of finding equations of straight lines through points – treating the vectors as position vectors. All methods were generally successful. The most common error was to find a scale factor but then apply it incorrectly. A quick check would have told learners that p was not equal to -2.4, which was the most common incorrect value.

Part (b)

Part (b) proved more challenging and learners had much less success, although many were able to make significant progress with the question. Where only the first mark was achieved, this was generally for correctly attempting $|d|$ as it was common for $|a| + |c|$ to be attempted in place of $|a + c|$. Learners who applied the modulus correctly to both sides of the equation were generally successful in achieving the next 2 marks as slips on squaring the 2 or even not including the 2 at all were condoned, with most forming a three-term quadratic in *q* and solving using an appropriate method. The most consistently successful attempts arose for learners who wrote $|d|$ as $6\sqrt{26}$ and divided both sides by 2 before removing the square root.

The alternative method, working with the modulus and noting that because the **j** components are of equal magnitude then the **i** components must also be of equal magnitude, was rarely successfully seen. A significant proportion of learners simply equated the **i** components which fortuitously lead to one correct value of q. This, alone, did not provide the evidence required to score marks as a correct method could not be implied without the correct modulus notation being used.

Question 7

This question, assessing index work, differentiation and integration, was attempted by the majority of learners and included many restart opportunities for those learners making errors. There were also follow through marks available for consistently accurate calculus of functions involving fractional indices.

Part (a)

In part (a), multiplying out the brackets and dividing by $\sqrt{x}$ was completed correctly by the majority of learners. For those who did not gain all the available marks in this part, there were a number of errors that were made, with the most common being the omission of the factor of 2 before the brackets. Others made sign errors and others missed an x in one or more terms, writing, for example, $50^{\frac{1}{2}}$ rather than $50x^{\frac{1}{2}}$. Other learners attempted to incorrectly ‘simplify’ their answer by dividing each coefficient by 2 and a small number of learners multiplied their quadratic by $\sqrt{x}$ instead of dividing.

Part (b)

In part (b), the majority of learners were able to demonstrate correct integration by correctly adding 1 to the power and dividing by the new power. The most common errors seen were incorrect coefficients arising from multiplication (than division) by the new power. Furthermore, the $+\ c$ was often missed. Learners who had made errors in (a) were able to access two out of the three marks in (b) provided their expression in (a) had fractional coefficients.

Part (c)

Part (c) was more challenging than part (b) and was less well attempted. Nonetheless, there were many learners who managed differentiation of the expression from (a) correctly and proceeded to complete the question.

However, in part (i), it was clear that some learners did not realise that differentiation of $f(x)$ was required and instead, differentiated the result from (b) or even the given equation in (c), neither of which could gain credit. Correctly differentiating an incorrectly simplified $f(x)$ of the required form, with fractional indices, could access two out of the three marks in this part. The demand in part (c) also involved manipulation of the derivative $f'(x) = 0$ to achieve the printed answer, which could be achieved in a single step but many learners did not gain the marks because they did not include the '$= 0$' until the final line. Others stated incorrectly that $x = x^{\frac{1}{2}}$ or similar before writing down the required equation.

In part (ii) many learners realised they needed to solve the given quadratic, but a significant proportion gave both solutions as their answer and did not realise that $x = -\frac{5}{3}$ should be discarded as the question stated that $x > 0$.

Question 8

This question assessed learners' use of trigonometric identities, their ability to solve trigonometric equations and use of trigonometric functions to model real-life scenarios. It proved well to differentiate between learners, with some marks of lower demand, sensible restart opportunities, and some more challenging marks, particularly in the accuracy in parts (i) and (ii)(b).

Part (i)

In part (i), most learners used $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and then multiplied by $\cos\theta$. However, a significant minority did not make much progress after this. A small number used an incorrect identities $\tan\theta = \frac{\cos\theta}{\sin\theta}$ or $\sin\theta = 1 - \cos\theta$ or used $2\tan\theta = \frac{2\sin\theta}{2\cos\theta}$. Others did not cancel or factor out $\sin\theta$. Better responses used $\cos^2\theta = 1 - \sin^2\theta$ and achieved an equation in $\sin^3\theta$ first, usually going on to solve an equation of the form $\cos^2\theta = k$ or $\sin^2\theta = k$. In the majority of cases, learners found one or two of the required solutions, with the $180°$ coming from $\sin\theta = 0$ the most frequently omitted. In this case it was either because they had cancelled the $\sin\theta$ or because they had $\sin\theta = 0$, found $\theta = 0$ but not the next value of $180°$.

Part (ii)

In part (b), the majority of learners substituted $t = 0$ into the model and achieved the correct answer including the units. A few learners did not include the units or did not evaluate $40 \sin 2°$ and did not gain the accuracy mark.

Part (c) proved to be the most demanding part of this question, and some learners made no attempt at this part. Those that did generally substituted $H = 25$ and proceeded to $\sin(0.3T + 2) = k$ before finding a correct value for $0.3T + 2$, which was usually 38.7. Typically, learners then solved to find T but either made no attempt to find the second smallest time, leaving their answer as $T = 122$ or used an incorrect method and only scored the first 2 marks. A minority of learners went on correctly to the answer of 464. Responses trying to solve the equation by attempting $\sin(0.3t + 2) = \sin 0.3t + \sin 2$ were reasonably rare.

Question 9

This question assessed learners' understanding of reciprocal graphs, translations, the discriminant and quadratic inequalities. The mixture of topics assessed here meant the question proved to differentiate well across the grade ranges and also gave learners access to later marks due to the restart opportunity in part (b).

Part (a)

Part (a) was attempted by almost all learners. Many were able to sketch the graph successfully, though a number attempted to plot points and often did not find one branch of the graph. Latitude was given on sketching the asymptotic behaviour of the graph but this could have been improved in many cases. Graphs often did not approach asymptotes closely and sometimes curved away from them. The most frequent error was down to the shape of the graph as many translated $y = \frac{1}{x}$ rather than $y = -\frac{1}{x}$

Some learners correctly identified the y intercept or vertical asymptote consistent with their graph. To secure full marks, learners needed to ensure that both the equation of the vertical asymptote and the y intercept were clearly labelled on the graph.

Part (b)

Part (b) was also well attempted, however for some, a lack of precision resulted in marks not being gained with the most common errors being in the manipulation of the quadratic to collect like terms or not acknowledging a correct inequality before the given answer. Others understood the demand to set the curve and the line equal and then manipulate to a quadratic however did not appreciate the need for the discriminant. It was often the case that terms were all on one side of the equals sign but not collected together and in such cases learners who made no further progress were not able to imply the correct quadratic.

Part (c)

In part (c), learners were required to solve the quadratic inequality result from part (b). Most learners successfully found the roots of the given quadratic but many were unable to apply the correct inequalities and/or set notation.
A sketch of the quadratic may have proved useful to help learners identify that they required the outer region, while those who found the outer region often did not include the critical values which were required due to the inclusive inequality in the question.

Question 10

This question assessed learners' understanding of the equation of a circle and their ability to problem solve. Part (b), in particular, proved to be quite challenging and it was not uncommon to see learners try a variety of unsuccessful approaches.

Part (a)

Part (a), was accessible, and most learners scored all three marks, completing the square on both terms and deducing the centre and radius. Occasionally learners gave their radius as $\sqrt{\frac{169}{4}}$ and this unsimplified answer was not condoned for the accuracy mark, while sign errors were seen in the centre, but this was also uncommon.

Part (b)

Part (b) was significantly more challenging. It appeared that many learners had some idea of how to solve the problem but did not use their centre correctly, instead finding the gradient between $(0, 0)$ and their centre. This significant error usually meant that no marks were scored in part (b). Of those that realised that $y = 0$, most went on to correctly solve and find $x = 10$: it was rare that $x = -2$ was selected. These learners typically went on to find the gradient between their centre and their P, scoring the first method mark, and form an equation. Most remembered to find the negative reciprocal and used the gradient of the tangent in a valid equation for the the equation of the tangent, but some used the gradient of the radius and as a result scored 2 marks in this part. Learners that progressed to an equation of the tangent almost always maintained the accuracy and rearranged the equation to the required form. It was rare that the "$= 0$" was omitted or for non-integer coefficients to be left in the equation.

Question 11

This question assessed learners' understanding of the laws of logarithms, the factor theorem and algebraic division, and the structure of the question meant that learners were able to attempt later parts of the question despite potentially finding earlier parts challenging.

Part (a)

Part (a) was a standard question assessing the laws of logarithms, and was generally well answered, with many learners demonstrating a solid understanding of the laws of logarithms. Most were able to apply the key rules, such as the power law and the subtraction law, to progress effectively through the problem. The majority of learners gained the first mark for correctly applying the power law and many successfully used the subtraction rule to combine logarithms into a single expression for the second mark. The third and fourth marks proved more challenging: those that removed the logarithms prior to expanding the brackets were generally more successful, as errors in expansion and in dealing with the fractions caused accuracy errors for many learners.

Part (b)

Part (b) tested the factor theorem and was also well answered with most learners able to substitute $x = 1$ into the cubic and obtain 0. However, the conclusion that $(x - 1)$ is a factor of $f(x)$ was often missing or insufficient, resulting in B0.

Part (c)

Part (c) tested algebraic division, with learners required to divide the cubic by $(x - 1)$. The most common method seen was long division, however the grid method and inspection were also seen, with all methods generally proving successful, possibly because the coefficient of the cubic term was 1. The most common error in this part was due to learners solving the quadratic (and in some cases the cubic) on their calculator, which was not acceptable due to the strict calculator warning at the start of the question.

Part (d)

Part (d) proved the most challenging. Learners were required to link the cubic and the original logarithm equation. Only a minority of learners realised that the solutions to the logarithm equation needed to be greater than 4, as solutions less than 4 would result in attempting to find a logarithm of a negative number. Those learners that substituted both $x = 1$ and $x = 6$ into the logarithm equation were more likely to be successful as they realised that $x = 1$ was invalid. Some of these learners did not identify the $x = 6$ was the solution, however, and instead gave the value obtained when $x = 6$ is substituted into either side of the original equation, which was 5.

Question 12

This question assessed learners' understanding of differentiation and integration and included elements of problem solving. It was very accessible to many learners, but finding the area of region R in part (c) proved to be challenging and meant that a small proportion achieved full marks. While the majority were able to make progress in each part, identifying a clear strategy for finding the area in part (c) proved to be the main barrier to complete success.

Part (a)

Part (a) was generally answered well. Most learners correctly differentiated the given function to obtain the gradient function and then evaluated it at the required point. Once the gradient had been found, the majority were able to proceed successfully to obtain the equation of the tangent. A small number of learners attempted to work backwards from the required answer or did not substitute the given value of x into their derivative, but these responses were uncommon. Working was usually well presented and easy to follow.

Part (b)

Part (b) highlighted a common issue across the cohort: the need to provide a clear conclusion. Many learners successfully substituted the given value into the equation and obtained the required result but did not state explicitly that this showed the curve crossed the x-axis at the given point.

Part (c)

Part (c) was the most demanding part of the question. Many learners demonstrated sound integration skills and were able to evaluate definite integrals accurately. However, a large proportion struggled to identify the geometry of the region *R* and consequently selected inappropriate limits of integration. Successful learners typically annotated the given diagram or drew a clear sketch showing all relevant coordinates and intersection points. The most reliable approach was to calculate the area of the appropriate triangle and subtract the area under the curve between the relevant limits. Common errors included integrating over incorrect intervals, using $x = 0$ as a lower limit, or using the intersection point of the line and curve without making the necessary adjustments to account for the shape of the region. Some learners attempted to integrate the difference between the line and the curve but selected incorrect limits. Even where a correct method had been established, arithmetic and calculator errors occasionally prevented learners from obtaining the final answer.

Overall, learners who carefully analysed the geometry of the region before beginning their calculations were significantly more successful than those who moved directly to integration without first identifying the boundaries of the required area.

Question 13

This question, involving an exponential model, was attempted by most learners and proved to be one of the most demanding questions as expected at this stage in the paper. The majority incorrectly used $23{,}000$ in part (a) and $5{,}000$ in part (b), not appreciating the significance of the "thousands" in the description of V. This resulted in a maximum of 1 mark which was scored in part (b).

Part (a)

In part (a), learners who substituted $V = 23$ usually achieved $A = 2$ and wrote the complete model scoring both marks. Some learners incorrectly used $t = 1$ instead of $t = 0$, while a small number of learners assumed $e^0 = 0$. As is often the case, some learners did not gain the final accuracy mark as they did not write out the complete equation for the model.

Part (b)

In part (b), where learners used $5{,}000$, they gained a maximum of 1 mark; this was very common. Some learners substituted 5 or $5{,}000$ but did not rearrange to collect the constant and as a result scored no marks.

For those that started the question correctly, using $V = 23$ in part (a) and $V = 5$ in part (b), they made progress through the question. When solving the equation (having combined the correct values of 2 and 5), the vast majority of learners did not recognise that they had a quadratic in $e^{-0.1T}$. It was common that learners took ln of each expression individually, and as a result scored just the first method mark. Learners who did recognise the quadratic, were usually successful in solving the equation. The most common method was to use a substitution (usually x or y as $e^{-0.1T}$). This allowed them to achieve the required value of $\frac{1}{4}$ and solve to give the final answer of 13.86. A small number used their calculator to achieve the final results and did not show sufficient working for full marks due to the requirements of the question.

Question 14

This question assessed learners' understanding of proof by deduction and, as the final question on the paper proved to be challenging to a significant number of learners. There was a substantial number of learners who did not attempt the question, possibly because they were unsure of how to proceed or possibly due to time constraints. Of those that did attempt a solution, a large proportion were unable to access any marks as they either attempted a 'trial and error' approach, or incorrectly considered odd and even cases for n rather than engaging with the less common formulation here of numbers which are not multiples of 3.

Many of the learners who did consider numbers that are not multiples of 3 usually began by setting n equal to either $3k + 1$ or $3k + 2$ or $3k - 1$ or $3k - 2$. However, many of these learners only proved the result for one of these cases and did not realise that it was necessary to consider a second distinct case in order to cover all the natural numbers that are not multiples of 3. It was also common that full marks were not achieved due to errors in expansions, errors in factorisation (when taking 3 out as a factor), not giving conclusions or for use of e.g. $n = 3n + 1$, i.e. defining n in terms of n, which did not gain the final accuracy mark.

It was possible to take a different approach, for example, consideration of numbers that are not multiples of 6 but, in this case, it was necessary to consider the four distinct cases to cover all natural numbers and full attempts at this approach were not seen. The alternative approach on the mark scheme was not seen.